

Notes: Vives (RFS, 2014)

Strategic Complementarity, Fragility, and Regulation

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1 Big picture

- **Question.** How do balance-sheet structure, information precision, and market stress jointly determine financial fragility, and what combination of solvency and liquidity regulation should be used to control crisis risk?
- **Setting.** The paper develops a theoretical crisis model built around a binary run/no-run decision by a continuum of investors. The main application is a bank-run environment with wholesale short-term funding, fire-sale losses, and noisy information about asset fundamentals.
- **Core challenge.** Liquidity and solvency problems are intertwined. A crisis can come from weak fundamentals, from self-fulfilling runs, or from the interaction of both. A regulator therefore needs a framework that links information, balance sheets, and investor coordination.
- **Main conceptual contribution.** The paper identifies the **degree of strategic complementarity** among investors' actions as the key sufficient concept for equilibrium selection, fragility, and policy. Fragility is high when small parameter changes move the equilibrium a lot or create discrete jumps through multiplicity.

- **Main mechanisms.**
 - A weaker balance sheet (higher short-term leverage, lower liquid reserves) makes a run more likely.
 - Greater funding pressure and larger fire-sale penalties make investors' actions more mutually reinforcing.
 - More precise *public* information can destabilize the system when fundamentals are weak because it helps investors coordinate on a run.
- **Main equilibrium result.** The crisis threshold is pinned down by a signal threshold s^* and a fundamental threshold θ^* . Crises occur when fundamentals fall below θ^* . The difference between the solvency threshold $\underline{\theta}$ and the equilibrium crisis threshold θ^* is the range of illiquidity driven by coordination failure.
- **Main comparative-static result.** Failure risk rises with higher short-term leverage, higher funding costs, higher fire-sale penalties, and more conservative investors. Better fundamentals reduce crisis risk. Public precision has a nontrivial multiplier effect.
- **Main policy result.** A solvency ratio and a liquidity ratio are both needed. They are **partial substitutes**: the solvency ratio is better for controlling insolvency, while the liquidity ratio is better for controlling illiquidity.
- **Key regulatory comparative statics.** More competition for short-term funds calls for a **tighter solvency requirement**. Higher fire-sale penalties or more conservative investors call for a **tighter liquidity requirement**. Stronger disclosure may also require tighter liquidity regulation.
- **Interpretive application.** The framework helps explain the 2007 run on structured investment vehicles (SIVs) and ABCP conduits, where the ABX index acted like a public signal that coordinated withdrawals in a highly leveraged and opaque environment.
- **Economic takeaway.** Disclosure, liquidity regulation, solvency regulation, and competition policy cannot be designed piecemeal. They interact because all of them change strategic complementarity and, through it, systemic fragility.

2 Model environment and key objects

2.1 Why this framework is useful

- The paper is designed to bridge the classic *panic* view of crises and the *fundamentals* view.
- It nests several well-known applications:
 - bank runs;
 - sovereign debt runs;
 - currency attacks;
 - credit market freezes.

- The leading example is a financial intermediary funded with unsecured short-term debt, liquid reserves, and stable funds.

Interpretation. The same coordination logic appears in several environments, but the paper's main bank-run setup is where the regulatory lessons are sharpest.

2.2 The investors' game

Each investor i chooses whether to *act* ($y_i = 1$) or *not act* ($y_i = 0$). In the bank-run interpretation, acting means selling, withdrawing, or refusing to roll over short-term debt.

Let y denote the mass of investors who act, and let θ denote the state of fundamentals. The payoff difference between acting and not acting is

$$\pi_1 - \pi_0 = \begin{cases} B > 0, & \text{if } y \geq h(\theta), \\ -C < 0, & \text{if } y < h(\theta), \end{cases} \quad (1)$$

where $h(\theta)$ is the **resistance function**: the minimum mass of investors that must act for the collective action to succeed.

Define

$$\gamma \equiv \frac{C}{B+C} \in (0, 1). \quad (2)$$

Interpretation. γ is the critical success probability that makes an investor indifferent between running and staying. More conservative investors have lower tolerance for failure, which in the paper is captured by a lower γ (or a higher γ^{-1}).

2.3 Complete information: solvency and supersolvency

Under complete information:

- if $\theta < \underline{\theta}$, acting is dominant and the institution fails even if nobody else runs;
- if $\theta \geq \bar{\theta}$, not acting is dominant and the institution survives even if everyone runs;
- if $\theta \in [\underline{\theta}, \bar{\theta})$, there are multiple equilibria.

Interpretation. The interval $[\underline{\theta}, \bar{\theta})$ is the region where coordination matters. The lower endpoint is a solvency threshold; the upper endpoint is a supersolvency threshold.

2.4 Information structure

Under incomplete information, investors have a Gaussian prior

$$\theta \sim N(\mu_\theta, \tau_\theta^{-1}), \quad (3)$$

and each investor receives a private signal

$$s_i = \theta + \varepsilon_i, \quad \varepsilon_i \sim N(0, \tau_\varepsilon^{-1}). \quad (4)$$

The parameters have natural interpretations:

- μ_θ : public assessment of fundamentals;
- τ_θ : precision of the public signal or prior;
- τ_ε : precision of private information.

Interpretation. The paper's most important informational insight is that better public information does not automatically stabilize. If fundamentals are weak, precise public information can make investors act more in unison and thereby increase fragility.

2.5 Bank-run balance sheet

At $t = 0$, the intermediary has:

- stable funds E (equity, long-term debt, insured deposits),
- short-term uninsured debt D_0 ,
- liquid reserves M ,
- risky investment I .

The balance-sheet identity is

$$E + D_0 = I + M. \quad (5)$$

At $t = 2$, the risky asset pays θI . Short-term creditors can demand payment of face value D at $t = 1$. Define:

$$m \equiv \frac{M}{D}, \quad \ell \equiv \frac{D}{E}, \quad d \equiv \frac{D}{D_0}. \quad (6)$$

- m is the **liquidity ratio**;
- ℓ is the **short-term leverage ratio**;
- d is the gross return promised on short-term debt and captures funding pressure / competition for funds.

Interpretation. The regulator's two main policy instruments are exactly these balance-sheet ratios: liquidity m and a leverage or solvency ratio involving $\ell^{-1} = E/D$.

2.6 Solvency threshold, supersolvency threshold, and fire sales

If no one runs ($y = 0$), the bank fails only if fundamentals are low enough:

$$\underline{\theta} \equiv \frac{D - M}{I} = \frac{1 - m}{\ell^{-1} + d^{-1} - m}. \quad (7)$$

If some investors run, the bank may need to liquidate assets early and incur a fire-sale penalty $\lambda > 0$. The **supersolvency threshold** is

$$\bar{\theta} \equiv (1 + \lambda)\underline{\theta}. \quad (8)$$

Interpretation. $\underline{\theta}$ is the level below which the intermediary is fundamentally insolvent. $\bar{\theta}$ is the level above which it is so strong that even a full run cannot kill it.

2.7 Resistance function in the bank-run model

For $\theta \geq \underline{\theta}$, the bank fails when too many investors withdraw, which can be written as

$$y > h(\theta) = m + \frac{\ell^{-1} + d^{-1} - m}{\lambda} (\theta - \underline{\theta}). \quad (9)$$

For $\theta < \underline{\theta}$, $h(\theta) < 0$, so failure is unavoidable.

Interpretation. The resistance function tells us how many withdrawals the bank can absorb at each fundamental level. A flatter $h(\theta)$ means fundamentals matter less for whether a run succeeds, so investors react more to each other and strategic complementarity becomes stronger.

2.8 Insolvency, illiquidity, and coordination failure

Under incomplete information, the equilibrium crisis threshold is θ^* , where $\theta^* \in [\underline{\theta}, \bar{\theta})$. Then:

- insolvency risk is $\Pr(\theta < \underline{\theta})$;
- overall failure risk is $\Pr(\theta < \theta^*)$;
- the range of illiquidity is $[\underline{\theta}, \theta^*)$.

Interpretation. In the interval $[\underline{\theta}, \theta^*)$, the institution is solvent but fails because investors do not coordinate on rolling over their claims. This is the paper's exact way of separating liquidity failures from solvency failures.

3 Models and analytical specifications (all equations + interpretation)

3.1 Equilibrium characterization

The equilibrium is characterized by two thresholds:

- a signal threshold s^* : investors act if and only if $s_i < s^*$;
- a failure threshold θ^* : the acting mass is successful if and only if $\theta < \theta^*$.

The two threshold conditions are:

$$\Pr(s < \hat{s} \mid \hat{\theta}) = h(\hat{\theta}), \quad (10)$$

and

$$\Pr(\theta < \hat{\theta} \mid \hat{s}) = \gamma. \quad (11)$$

The paper writes the best response in threshold space as

$$r(\hat{s}) = \frac{\tau_\theta + \tau_\varepsilon}{\tau_\varepsilon} \theta_F(\hat{s}) - \frac{\tau_\theta}{\tau_\varepsilon} \mu_\theta - \frac{\sqrt{\tau_\theta + \tau_\varepsilon}}{\tau_\varepsilon} \Phi^{-1}(\gamma), \quad (12)$$

where $\theta_F(\hat{s})$ is the implied failure threshold if others use signal threshold $\hat{s}$, and $\Phi(\cdot)$ is the standard normal cdf.

Interpretation. The equilibrium is a fixed point of the best response. Investors choose a signal cutoff based on what they think others' cutoff will be.

3.2 Proposition 1: existence, multiplicity, and uniqueness

The paper's equilibrium characterization has three parts.

- An equilibrium is described by (s^*, θ^*) , and the crisis probability conditional on $s = s^*$ equals γ .
- There exists a critical level $\bar{h}_0(\underline{\theta})$ such that if $h(\underline{\theta})$ is high enough, the smallest equilibrium is just the no-run threshold $\theta^* = \underline{\theta}$.
- The equilibrium is unique if

$$\frac{\tau_\theta}{\sqrt{\tau_\varepsilon}} \leq h_1 \sqrt{2\pi}, \quad (13)$$

where h_1 is the smallest slope of the resistance function on $(\underline{\theta}, \bar{\theta})$.

Interpretation. Multiplicity needs sufficiently strong strategic complementarity. If public information is too precise relative to private information, or if the resistance function is too flat, multiplicity becomes more likely.

3.3 Strategic complementarity as the core fragility measure

The slope of the best response is the paper's measure of strategic complementarity. Its maximal value is

$$\bar{r}' \equiv \frac{\tau_\theta + \tau_\varepsilon}{\tau_\varepsilon + h_1 \sqrt{2\pi\tau_\varepsilon}}. \quad (14)$$

Strategic complementarity is stronger when:

- h_1 is smaller, so the resistance function is less sensitive to fundamentals;
- τ_θ is larger, so public information is more precise;
- τ_ε is small enough relative to τ_θ , so private signals do not generate much dispersion in beliefs.

In the bank-run case, this means stronger complementarity with:

- higher short-term leverage ℓ ,
- higher promised return on debt d ,
- larger fire-sale penalty λ ,
- and, in the empirically relevant parameter range, a higher liquidity ratio m , because it flattens the resistance function.

Interpretation. The paper's main insight is that fragility is not only about weak fundamentals. It is about how much investors' actions become strategic complements.

3.4 Proposition 2: comparative statics of crisis risk

The paper parameterizes stress by a variable α such that $\partial h / \partial \alpha < 0$: more stress lowers the mass of withdrawals needed for a run to succeed.

Proposition 2 says that, at extremal equilibria or under adaptive dynamics:

- θ^* , s^* , and $\Pr(\theta < \theta^*)$ are decreasing in γ and in μ_θ , but increasing in stress α ;
- a public signal has a **multiplier effect** on equilibrium thresholds beyond its direct effect on beliefs, and this multiplier is stronger when τ_θ is higher;
- if fundamentals are weak enough (a sufficient condition is $\mu_\theta \leq \underline{\theta}$) and $\gamma < 1/2$, then more precise public information raises θ^* , the crisis probability, and the range of illiquidity, while more precise private information lowers them.

Interpretation. With weak fundamentals, public precision is dangerous because it coordinates action. With strong fundamentals, the effect reverses.

3.5 Bank-run comparative statics: insolvency versus failure

The paper summarizes the bank-run comparative statics for the case $m < m^\dagger$, where $m^\dagger$ is a critical liquidity ratio above which the smallest equilibrium coincides with $\underline{\theta}$.

	$\Pr(\theta < \underline{\theta})$ (insolvency)	$\Pr(\theta < \theta^*)$ (failure)	$\theta^* - \underline{\theta}$ (range of illiquidity)
m (liquidity ratio)	—	—	—
μ_θ (strength of fundamentals)	—	—	—
ℓ (leverage ratio)	+	+	+
d (cost of funds)	+	+	+
λ (fire-sale penalty)	0	+	+
γ^{-1} (investor conservatism)	0	+	+

Table 1: Table 1: comparative statics of insolvency, failure, and illiquidity in the bank-run model for $m < m^\dagger$.

Interpretation.

- Insolvency depends on the balance sheet directly, so it is affected by m, μ_θ, ℓ, d , but not by λ or γ .
- Failure and illiquidity depend on strategic interaction as well, so they also react to λ and γ .

3.6 Weak fundamentals, transparency, and bad news

The paper's most important information result is that public precision can be destabilizing when fundamentals are weak.

- Figure 4 shows that increasing τ_θ rotates the best response upward and can move the equilibrium toward the run outcome.

- Figure 5 shows that the same bad-news shock has a much larger effect when public precision is high.

Interpretation. In stressed and opaque environments, more disclosure can make a bad public signal much more powerful as a coordinating device.

3.7 The regulator's problem

The regulator wants to cap:

- the probability of insolvency at q ,
- the probability of overall failure at p .

These goals imply two constraints:

$$\underline{\theta} \leq \underline{\theta}_q \equiv \mu_\theta + \Phi^{-1}(q)/\sqrt{\tau_\theta} \quad (\text{solvency constraint S}), \quad (15)$$

and

$$\theta^* \leq \theta_p^* \equiv \mu_\theta + \Phi^{-1}(p)/\sqrt{\tau_\theta} \quad (\text{liquidity constraint L}). \quad (16)$$

Interpretation. The solvency constraint controls the probability that the intermediary is fundamentally insolvent. The liquidity constraint controls the probability that it fails for *any* reason, including runs.

3.8 Proposition 3: minimal liquidity and solvency ratios

The paper solves for the minimal regulatory ratios $\hat{m}$ and $\hat{\ell}^{-1}$ that satisfy both constraints. They are:

$$\hat{m} = \max \left\{ 1 - (1 - k) \left(1 - \lambda^{-1} \left(\frac{\theta_p^*}{\underline{\theta}_q} - 1 \right) \right)^{-1}, 0 \right\}, \quad (17)$$

and

$$\hat{\ell}^{-1} = (\underline{\theta}_q)^{-1} - d^{-1} - [(\underline{\theta}_q)^{-1} - 1] \hat{m}. \quad (18)$$

Here $k \in (0, 1)$ rises with the regulator's tolerance p , with investor conservatism γ^{-1} , and, in the relevant case, with public precision. When $\tau_\varepsilon \rightarrow \infty$, $k = 1 - \gamma$.

The paper also shows:

- if $p = q$, then $\hat{m} = k$;
- if fire-sale penalties are negligible, the liquidity requirement can collapse to zero;
- when fire-sale penalties are significant and investors are conservative, a positive liquidity requirement is necessary.

Interpretation. Liquidity regulation matters exactly when solvent institutions are still vulnerable to run-induced liquidation losses.

3.9 Regulatory comparative statics

When $p > q$ and the liquidity constraint binds, the paper obtains the following sign pattern.

	λ fire-sale penalty	γ^{-1} investor conservatism	d cost of funds	μ_θ strength of fundamentals	τ_θ public precision
$\hat{m}$ (liquidity ratio)	+	+	0	+	+
$\hat{\ell}^{-1}$ (solvency ratio)	−	−	+	−	−

Table 2: Table 2: comparative statics of the minimal regulatory ratios when the liquidity constraint binds. * Requires $p < 1/2$ and either $\tau_\varepsilon \rightarrow \infty$ or τ_ε small with $p > \gamma$.

Interpretation.

- Higher **funding pressure** d calls for tighter solvency regulation but leaves the liquidity requirement unchanged.
- Higher **fire-sale penalties** λ or more conservative investors call for tighter liquidity requirements and looser solvency requirements.
- Higher **public precision** may require tighter liquidity regulation and looser solvency regulation, so prudential and disclosure policy must be designed together.

3.10 Why competition, disclosure, and prudential policy interact

The paper is very explicit about three policy interactions.

- **Competition policy and solvency.** If competition for short-term funds increases d , the regulator should strengthen solvency requirements.
- **Disclosure and liquidity.** If public signals become more precise, for example because of a derivatives market or new disclosure rules, liquidity requirements may need to be tightened.
- **Stress and policy mix.** In turbulent environments with high fire-sale penalties and conservative investors, the regulator should rely relatively more on liquidity requirements and relatively less on solvency requirements.

Interpretation. A one-dimensional capital rule is not enough. The right policy mix depends on what type of fragility is most likely.

3.11 Interpretation of the 2007 run on structured investment vehicles

The paper maps the model to the 2007 SIV and ABCP run as follows:

- At $t = 0$, mortgages are originated and securitized.
- An SIV holds risky mortgage-backed assets I and reserves M , financed by equity E and short-term debt D_0 .

- At $t = 1/2$, a public signal about fundamentals is released; in the paper’s interpretation, this can be the ABX index.
- At $t = 1$, short-term fund managers decide whether to roll over funding.
- At $t = 2$, mortgage-backed assets pay off.

The paper argues that ABX and CMBX became highly visible public signals. As bad news on subprime mortgages accumulated, these indexes fell sharply, and the run on ABCP conduits and SIVs unfolded through non-rollover of short-maturity paper.

Interpretation. The SIV episode fits the model because leverage was high, fire-sale losses were severe, and public bad news from ABX helped investors coordinate on withdrawal.

3.12 Other applications

The framework also applies to:

- **Sovereign debt runs.** The bank-run model can be reinterpreted with reserves M , short-term external debt D_0 , and a public signal such as a sovereign rating. The same logic yields reserve requirements and external leverage constraints.
- **Currency attacks.** In a simplified Morris–Shin style setup, the resistance function is $h(\theta; \alpha) = \alpha^{-1}\theta$. The crisis probability falls with stronger reserves and higher attack costs, but rises with stress.
- **Credit market freezes.** In the Bebchuk–Goldstein style application, $h(\theta; \alpha) = \alpha(\theta - \underline{\theta}(\alpha))$. Interestingly, higher stress may reduce strategic complementarity there, showing that the mapping from stress to coordination can differ across environments.

Interpretation. The paper is not only about banks. It offers a common language for crises where investors’ actions become mutually reinforcing.

4 Identification and analytical credibility (what makes the results persuasive)

4.1 Why strategic complementarity is the right organizing concept

The paper’s main analytical move is to summarize fragility through the slope of the best response.

- Balance-sheet stress changes the resistance function.
- Information precision changes how strongly investors react to a common signal.
- Market stress changes the gain from coordinating on a run.

Interpretation. Once these channels are translated into strategic complementarity, the comparative statics become much more transparent.

4.2 What the model can separate clearly

The model sharply distinguishes:

- **solvency** ($\theta < \underline{\theta}$);
- **run-induced failure** ($\underline{\theta} \leq \theta < \theta^*$);
- **supersolvency** ($\theta \geq \bar{\theta}$).

The strategic behavior of investors does *not* affect $\underline{\theta}$, but it *does* affect θ^* .

Interpretation. This is why the paper can say something precise about when a bank is solvent but still fails for liquidity reasons.

4.3 Why disclosure can destabilize

The paper's result on disclosure is not a blanket anti-transparency claim. It is conditional:

- if fundamentals are weak;
- if public signals are strong relative to private signals;
- if assets are opaque enough that investors look to the public signal for coordination.

Interpretation. Transparency helps when it improves beliefs without creating a focal point for panic. It hurts when it mainly coordinates withdrawals.

4.4 Why the solvency and liquidity ratios are only partial substitutes

A stronger solvency ratio can indirectly lower crisis risk, and a stronger liquidity ratio can indirectly reduce insolvency risk through equilibrium effects, but they do not do the same job.

- The solvency ratio mainly shifts $\underline{\theta}$.
- The liquidity ratio mainly shifts θ^* relative to $\underline{\theta}$.

Interpretation. This is why the regulator generally wants both instruments, not one or the other.

4.5 Limits of the model

The paper is explicit about its limitations.

- The intermediary's balance sheet and the regulator's objectives are exogenous.
- The model is basically static, even though it is meant to speak to dynamic crisis episodes.
- Investors are symmetric.
- The analysis focuses on a single institution or a consolidated banking sector, so it does not model contagion across banks.
- Market parameters such as the fire-sale penalty are taken as given.

Interpretation. The framework is best read as a clean comparative-static laboratory, not a full quantitative macro-financial model.

4.6 Relation to the literature

The paper bridges and extends several literatures.

- It connects panic-based models such as Diamond–Dybvig with information-based global-games models.
- It nests Morris and Shin style currency attacks, Rochet–Vives bank runs, and Bebchuk–Goldstein credit freezes.
- It contributes to prudential-regulation debates by linking leverage, liquidity, and transparency in one framework.

Interpretation. The novelty is not any one application by itself, but the unified strategic-complementarity lens across all of them.

5 Tables and figures

5.1 Figure 1: the resistance function in the bank-run model

Read:

- Figure 1 plots the resistance function $h(\theta)$ against fundamentals.
- The intercept at $\underline{\theta}$ is the liquidity ratio m .
- A higher fire-sale penalty λ or higher promised return on short-term debt d rotates the function down, so fewer withdrawals are needed for failure.

Economic message. This figure is the paper’s simplest visual summary of fragility: weaker fundamentals, higher funding pressure, or larger fire-sale losses all make the institution easier to run.

5.2 Figures 2 and 3: equilibrium thresholds and multiplicity

Read:

- Figure 2 shows the equilibrium as the intersection of the signal-threshold curve and the failure-threshold curve.
- Figure 3 shows how the best response can cross the 45-degree line once or several times depending on the slope of the resistance function.
- As h_1 falls, strategic complementarity rises and multiplicity becomes possible.

Economic message. The paper’s fragility concept is visual here: a flatter resistance function creates steeper best responses and opens the door to multiple run equilibria.

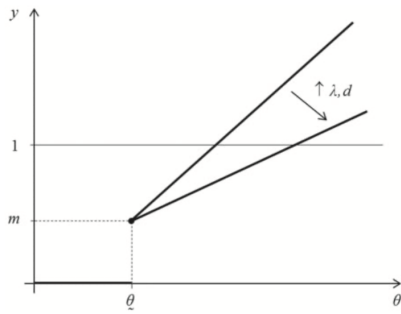


Figure 1
The resistance function in the bank runs model: $h(\theta) = m + (\ell^{-1} + d^{-1} - m)\lambda^{-1}(\theta - \theta)$ if $\theta \geq \theta$, and $h(\theta) < 0$ if $\theta < \theta$. Above the bold line $y > h(\theta)$ for $\theta \geq \theta$ and the bank fails.

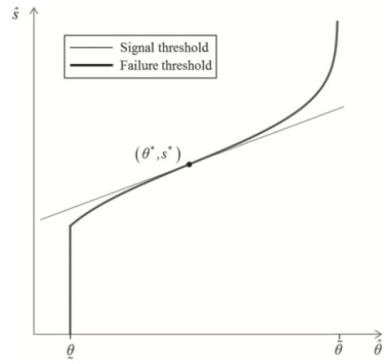


Figure 2
Equilibrium characterization in the bank run model
Parameters include $\gamma = 0.5$, $m = 0.1$, $\ell^{-1} = 0.2$, $d^{-1} = 0.9$, $\lambda = 1$, $\theta = 0.9$, $\theta = 1.8$, $\tau_\theta = 0.5$, $\tau_e = 0.05$, and $\mu_\theta = 1.3$.

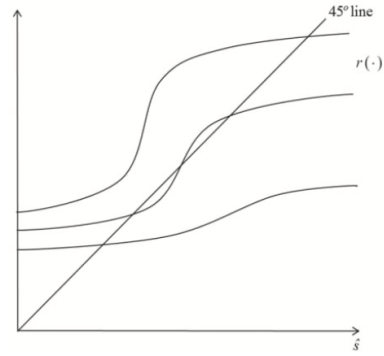


Figure 3
Possible best responses of a player to the threshold strategy $\hat{s}$ used by rivals for an intermediate range of γ . The lower (resp., middle, upper) plot corresponds to high (resp., intermediate, low) values of h_1 .

5.3 Figure 4 and Figure 5: public precision and bad news

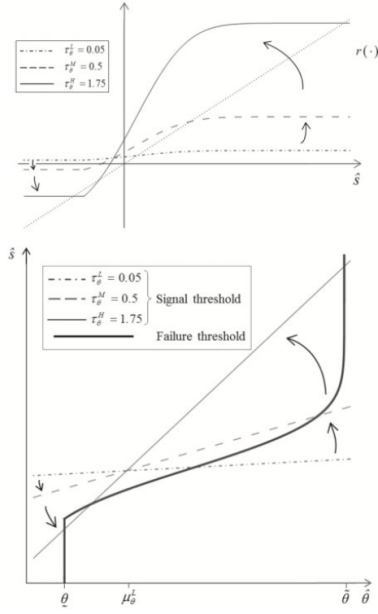


Figure 4
Increase in public precision ($\tau_\theta^L = 0.05$, $\tau_\theta^M = 0.5$, $\tau_\theta^H = 1.75$) with weak fundamentals (low public signal, $\mu_\theta^L = 1.1$)
Other parameters include $\tau_\varepsilon = 0.05$, $\gamma = 0.5$, $m = 0.1$, $\ell^{-1} = 0.2$, $d^{-1} = 0.9$, $\lambda = 1$, $\underline{\theta} = 0.9$, and $\bar{\theta} = 1.8$.

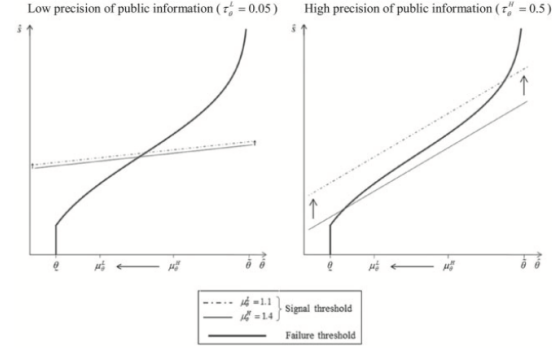


Figure 5
The effect of bad news (from $\mu_\theta^H = 1.4$ to $\mu_\theta^L = 1.1$)
Other parameters include $\tau_\varepsilon = 0.05$, $\gamma = 0.5$, $m = 0.1$, $\ell^{-1} = 0.2$, $d^{-1} = 0.9$, $\lambda = 1$, $\underline{\theta} = 0.9$, $\bar{\theta} = 1.8$, $\tau_\theta^L = 0.05$, and $\tau_\theta^H = 0.5$.

Read:

- Figure 4 shows that, with weak fundamentals, higher public precision can push the equilibrium toward the run outcome.
- Figure 5 shows that the same bad-news shock moves the crisis threshold far more when public precision is high.

Economic message. These figures are the cleanest illustration of the paper's most counterintuitive result: more precise public information can increase fragility when fundamentals are weak.

5.4 Table 1: comparative statics of insolvency and failure risk

Read:

- Table 1 separates effects on insolvency from effects on overall failure and the range of illiquidity.
- Liquidity m and stronger fundamentals μ_θ reduce all three.
- Higher leverage ℓ and higher funding pressure d raise all three.
- Fire-sale penalties λ and investor conservatism matter only for failure and illiquidity, not for insolvency itself.

Economic message. The table is important because it shows that some parameters work through fundamental solvency, while others work mainly through coordination.

5.5 Table 2 and Figure 7: prudential policy mix

Table 1
Comparative statics of solvency and liquidity risk for $\alpha < \bar{\alpha}$

	$\Pr(\theta < \theta^*) \mid \text{Solvency}$	$\Pr(\theta < \theta^*) \mid \text{Failure}$	$\theta^* - \bar{\theta}$ (Range of liquidity)
m (Liquidity ratio)	(-)	(-)	(-)
μ_θ (Strength of fundamentals)	(-)	(-)	(-)
$\bar{\theta}$ (Leverage ratio)	(+)	(+)	(+)
d (Cost of funds)	(+)	(+)	(+)
λ (Fire-sale penalty)	(0)	(+)	(+)
γ^{-1} (Investor conservatism)	(0)	(+)	(+)

Table 2
Comparative statics of regulatory ratios when $p > q$ and $\lambda k > (\theta_p^*/\theta_q) - 1$

	$\hat{\lambda}$ (Fire-sale penalty)	γ^{-1} (Investor conservatism)	$\bar{d}$ (Cost of funds)	μ_θ (Strength of fundamentals)	τ_θ (Precision of public information)
$\hat{m}$ (Liquidity ratio)	(+)	(+)	0	(+)	(+)*
$\hat{\ell}^{-1}$ (Solvency ratio)	(-)	(-)	(+)	(-)	(-)*

* Provided $p < 1/2$ and either $\tau_\theta \rightarrow \infty$ or τ_θ is small and $p > \gamma$.

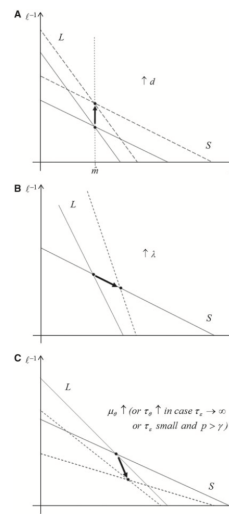


Figure 7
A. Effects of an increase in the cost of funds. B. Effects of an increase in the fire-sale penalty. C. Effects of an increase in the strength of fundamentals (good news) or in the precision of public information (transparency) under certain conditions.

Read:

- Table 2 shows that the liquidity requirement $\hat{m}$ and the solvency requirement $\hat{\ell}^{-1}$ often move in opposite directions.
- Figure 7 visualizes three cases:
 - higher funding costs rotate the solvency and liquidity constraints in a way that tightens the solvency requirement;
 - higher fire-sale penalties tighten the liquidity requirement;
 - stronger fundamentals or more precise public information can also tilt the optimal mix toward liquidity regulation.

Economic message. The regulator should not ask only “more capital or less capital?” The right answer depends on whether the stress comes from leverage, funding competition, fire-sale risk, or disclosure.

5.6 Figure 9: the ABX collapse and the SIV run

Read:

- Figure 9 plots the BBB tranche of several ABX vintages.
- The indexes are near par through much of 2006, then fall sharply during 2007.
- The paper interprets this as the visible public signal that helped coordinate the run on SIVs and ABCP conduits.

Economic message. Figure 9 is where the paper connects theory to the crisis narrative: a transparent market-based signal can become the focal point for panic in a leveraged system.

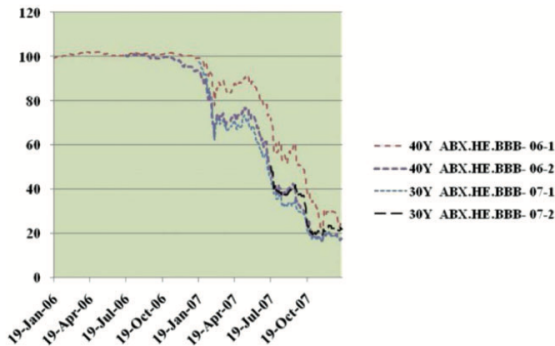


Figure 9
Prices of the 2006:1, 2006:2, 2007:1, and 2007:2 vintages of the ABX index for the BBB tranche

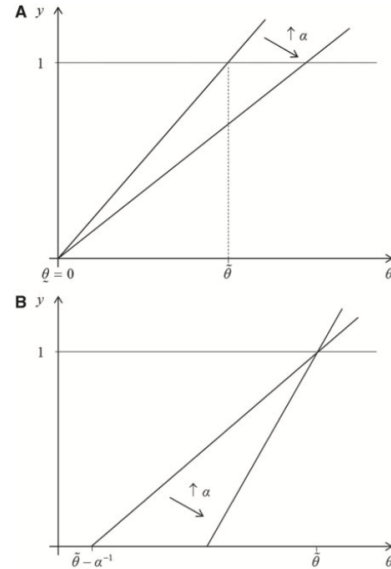


Figure 10
A. The resistance function $h(\theta; \alpha) = \alpha^{-1}\theta$ in the model of currency crises. B. The resistance function $h(\theta; \alpha) = \alpha(\theta - \bar{\theta}(\alpha))$ in the model of credit market freezes

5.7 Figure 10: currency attacks and credit freezes

Read:

- Figure 10A shows the resistance function for a currency-attack application.
- Figure 10B shows the resistance function for a credit-freeze application.
- In the credit-freeze case, more stress can reduce strategic complementarity, which contrasts with the bank-run case.

Economic message. The general framework is portable, but the way stress maps into strategic complementarity depends on the specific environment.

6 Short summary

- The paper studies crises through a global-games framework in which investors decide whether to run.
- The key object is the resistance function $h(\theta)$, which determines how many investors must act for a run to succeed.
- Fragility is governed by strategic complementarity: when investors react strongly to each other, small shocks can create large equilibrium changes.
- In the bank-run application, fragility rises with short-term leverage, funding pressure, fire-sale penalties, and precise public information.
- The crisis threshold θ^* exceeds the solvency threshold $\underline{\theta}$ when solvent institutions can still fail because of coordination.

- The regulator needs both a solvency ratio and a liquidity ratio.
- More competition for funding strengthens the case for tighter solvency regulation.
- Higher fire-sale penalties, more conservative investors, and sometimes more disclosure strengthen the case for tighter liquidity regulation.
- The model provides a coherent interpretation of the 2007 SIV run and also extends to sovereign debt runs, currency attacks, and credit freezes.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the central state variable for crisis analysis is not just fundamentals but the degree of strategic complementarity among investors' actions. Once strategic complementarity is linked to balance-sheet structure, market stress, and information precision, one can characterize both fragility and the right mix of solvency and liquidity regulation.

7.2 Economic intuition

A bad balance sheet or a strong public bad signal does not merely lower expected payoffs. It changes how much investors want to move *with* each other. That coordination channel is what turns ordinary stress into systemic fragility. Regulation works by shifting either the fundamental insolvency threshold or the run threshold, and the optimal policy depends on which margin is most dangerous.

7.3 Takeaway

The paper's deepest message is that prudential regulation, competition policy, and disclosure policy are interdependent. Higher transparency is not always stabilizing, higher competition is not harmless if leverage is unchanged, and a liquidity rule cannot replace a solvency rule one-for-one. The design of financial regulation must therefore account for how policy changes investor coordination incentives, not just balance-sheet arithmetic.